

PROJECT REPORT
ON
“Timetable Generator”

Bachelor of Computer Applications (BCA)

To

UTTARANCHAL SCHOOL OF COMPUTING SCIENCES

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Under The Supervision of

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Rohan Gurung

BCA IVF

DECLARATION

I hereby declare that the project report entitled “**TIMETABLE GENERATOR**” submitted by **JAI KARAN, Rohan Gurung Enrollment No. UU2409000144, UU2409000268** to Uttarakhand School of Computing Science. The project was done under the Guidance of **Mr. Deepak Kumar Pathak** I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this university or any other university or institute.

Rohan Gurung

BCA IV-F

CERTIFICATE

This is to certify that the project entitled “**TIMETABLE GENERATOR**” by **JAI KARAN and ROHAN GURUNG** Enrollment No. **UU2409000144,UU2409000268** and has been submitted in the partial fulfilment of the requirements for the award of the degree of BCA from Uttaranchal University, Dehradun. The results embodied in this project have not been submitted to any other University or Institution for the record of any degree.

Under the guidance of:

Mr. Deepak Kumar Pathak

Assistant professor,

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1. Introduction

The Timetable Generator System is a software-based solution developed to automate the complex and time-consuming process of timetable scheduling in educational institutions. In schools and colleges, preparing a timetable manually requires careful coordination of subjects, teachers, classrooms, and available time slots. Even a small mistake can lead to scheduling conflicts such as assigning the same teacher to multiple classes at the same time or allocating the same classroom for different lectures simultaneously. This project aims to eliminate such errors by providing an automated and efficient scheduling system.

In traditional systems, timetable preparation is usually handled manually by administrative staff, which consumes significant time and effort. Manual scheduling also makes it difficult to maintain fairness in workload distribution among teachers and proper spacing of subjects throughout the week. Additionally, making changes to the timetable due to sudden events, faculty leave, or classroom unavailability becomes complicated. The Time Table Generator System addresses these challenges by applying logical algorithms and structured programming techniques to generate a conflict-free and optimized schedule.

This system is developed using Java programming language and is based on core programming concepts such as object-oriented programming (OOP), arrays, loops, and conditional statements. The application collects essential input including number of classes, subjects, teachers, working days, and time periods. After processing these inputs, the system automatically assigns subjects and faculty members to available slots while ensuring that no overlapping or duplication occurs. The system follows predefined constraints to maintain accuracy and efficiency.

The primary goal of this project is not only to reduce human effort but also to improve the reliability and transparency of the scheduling process. By automating timetable generation, institutions can save valuable administrative time and minimize human errors. The system ensures equal subject distribution, proper teacher workload management, and optimal classroom utilization. It also provides flexibility for future modifications and enhancements.

2. Objectives

- To automate timetable generation.
- To avoid teacher and classroom conflicts.
- To ensure equal subject distribution.
- To reduce manual errors.
- To improve administrative efficiency.
- To provide an easy-to-use system for timetable management.

3. System Analysis

Manual timetable preparation consumes time and may result in scheduling conflicts. There is a need for an automated system that efficiently manages resources and ensures accuracy.

3.1.Existing System Problems

- Teacher time clashes
- Classroom conflicts
- Unequal subject allocation
- Time-consuming manual process

3.2.Proposed System Features

- Automated schedule generation
- Conflict detection system
- User-friendly interface
- Centralized timetable display

4. Project Planning

Phase 1 — Requirement Gathering

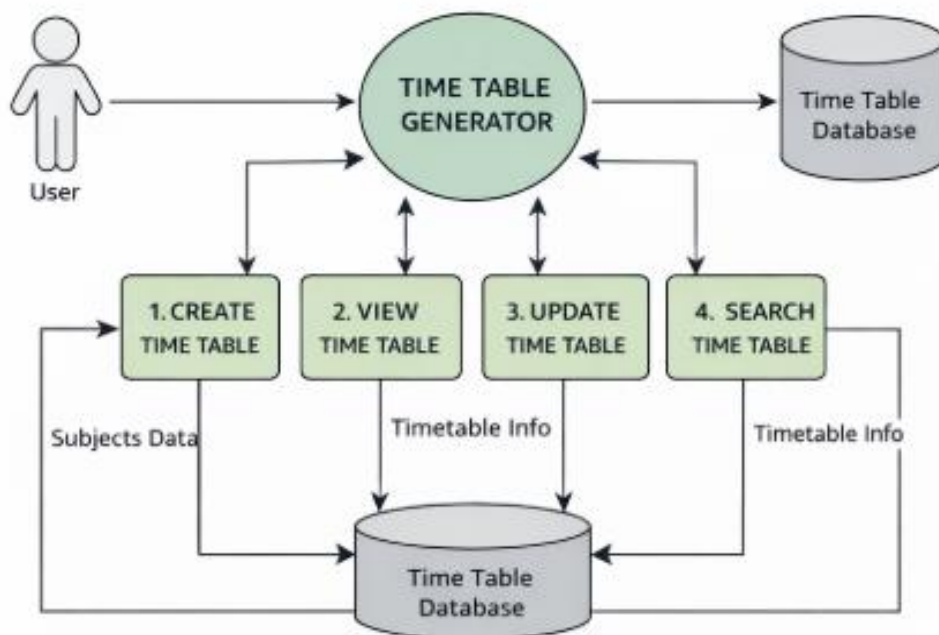
Phase 2 — System Design

Phase 3 — Implementation (Java)

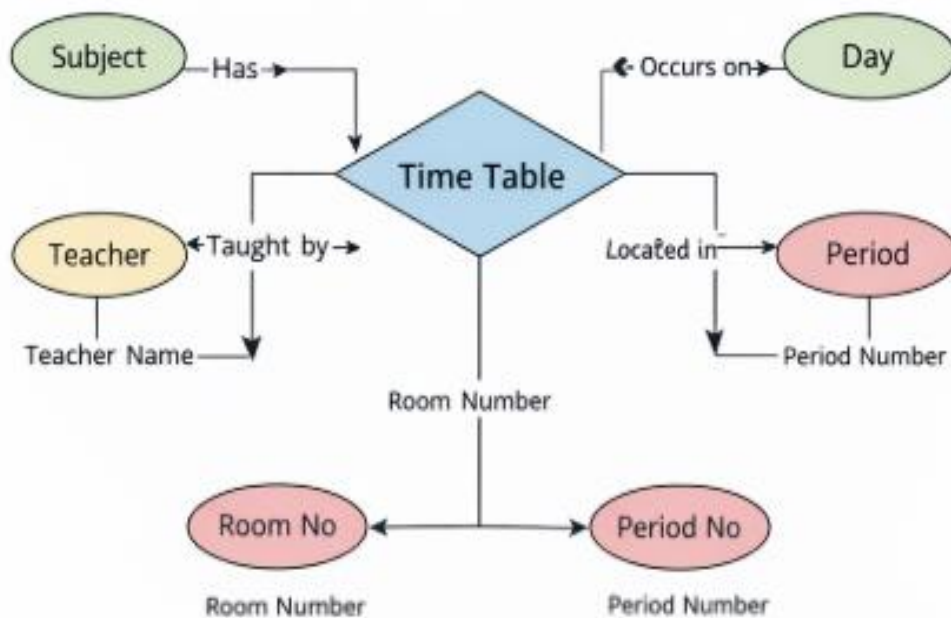
Phase 4 — Testing

Phase 5 — Documentation & Presentation

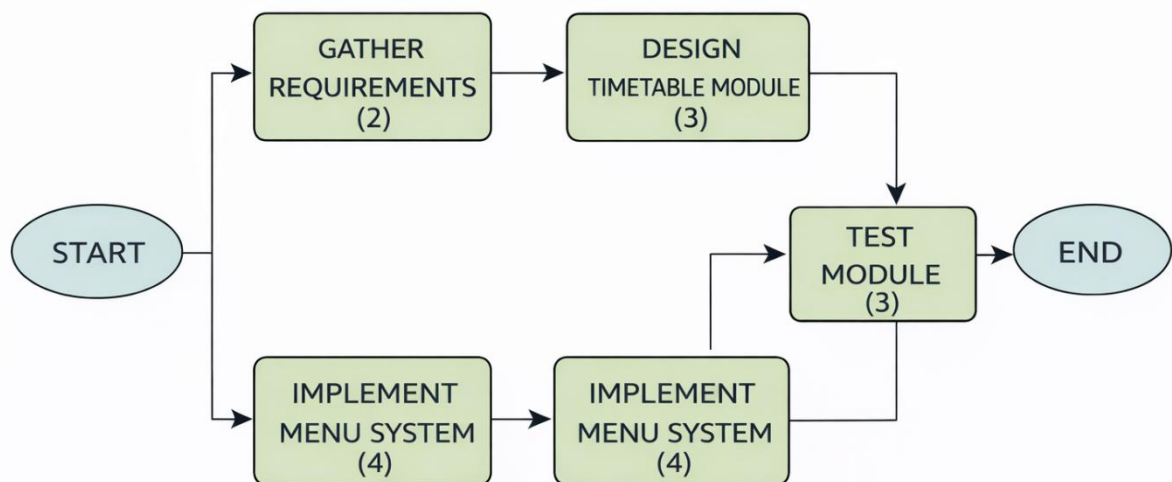
4.1.Data flow diagram



4.2.E-R diagram



4.3.Pert Chart



4.4.Gantt Chart

Task	Duration / Days (h)	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 10
Gather Requirements	2								
Design Timetable Module	3								
Develop Timetable Function	5								
Implement Menu System	4								
Test Module	3								

5. Requirement Analysis

Developing a Timetable Generation system begins by understanding the unique needs of educational institutions. Through discussions with stakeholders and careful data collection, we lay the groundwork for a system that simplifies timetable creation and boosts scheduling efficiency. This initial phase guides decisions on how the system will be built, ensuring it meets the specific demands of educational scheduling.

5.1. Stakeholder Requirements:

This Project involved Stakeholder Requirements for understanding the scheduling needs and preferences of educational institutions, administrators, teachers, and students. This input guides the development process to create a system that aligns with the diverse demands of the stakeholders. Following is the required constraints given by the user:

1.1 Administrators: • User-Friendly Interface: The Timetable Generator should have an intuitive and easy-to-use interface, allowing administrators to navigate through the system effortlessly. • Customization Options: Administrators should have the ability to customize the timetable creation process, specifying rules, preferences, and constraints.

1.2 End Users (Faculty and Students): • User Feedback Mechanism:

End users need a user-friendly feedback mechanism within the application to provide insights into their experiences with the schedules. End users desire a system that actively incorporates their feedback to continuously improve the scheduling algorithms and user experience.

5.2. Functional Requirements:

Functional requirements for the Timetable Generation system encompass the specific capabilities and features needed, such as efficient scheduling algorithms, real-time updates, and user-friendly interfaces. to ensure the system meets its intended functionalities, contributing to streamlined timetable creation and enhanced user experience.

5.2.1 Cross-Platform Application: • The Timetable Generator application should be developed using the Flutter framework to ensure a consistent and user-friendly experience

across iOS, Android, and Web platforms. • The interface should be responsive and adaptable to various screen sizes and resolutions.

• Following are the technologies involved in cross-platform application development:

(1) Flutter: Flutter is an open-source UI software development toolkit created by Google. It is used to build natively compiled applications for mobile, web, and desktop from a single codebase. It provides a reactive framework, expressive UI, and a set of pre-designed widgets.

Use Cases: • Flutter is used to build mobile applications for both iOS and Android. • It is increasingly used for building web and desktop applications as well.

(2) Dart: Dart is a programming language developed by Google and is the primary language used for building applications with Flutter. It is designed for building scalable and efficient web, mobile, and server applications.

Use Cases: • Dart is primarily used for building applications with Flutter. • It can be used for server-side development, web applications, and command-line tools. **(3) Android Studio:** Android Studio is the official integrated development environment (IDE) for Android app development. It is based on JetBrains' IntelliJ IDEA and tailored for Android development.

5.3. Non-Functional Requirements:

5.3.1 Usability: • The application should have an intuitive and user-friendly interface, ensuring that users can easily navigate and interact with the Timetable Generator.

- Provide clear and concise documentation and onboarding materials to assist users in understanding and using the application effectively.
- The system should generate timetables promptly, with a maximum response time of [define specific time] seconds to ensure a responsive user experience.
- The application should handle a concurrent user load of [define specific number] users without significant performance degradation.

5.3.2 Reliability:• The Smart Timetable Generator should have a high level of availability, with a target uptime of [define specific percentage] during normal operating hours.

- Implement robust error handling and recovery mechanisms to minimize disruptions in case of unexpected errors.

5.3.3 Security: • Ensure data confidentiality and integrity by implementing secure authentication and authorization mechanisms for user access.

- Protect against common security threats such as SQL injection, cross-site scripting (XSS), and data breaches.

5.3.4 Compatibility: • Ensure cross-browser compatibility, supporting the latest versions of popular web browsers (e.g., Chrome, Firefox, Safari).

- The mobile application should be compatible with the latest versions of iOS and Android operating systems.

5.3.5 Maintainability:

- Use modular and well-documented code to facilitate ease of maintenance and updates.
- Provide a mechanism for administrators to apply system updates and patches seamlessly.

6. Design And Architecture

In crafting the design and architecture for a Timetable Generation system, the focus is on creating a robust and efficient framework that seamlessly integrates scheduling algorithms, user interfaces, and database management. This involves careful consideration of scalability, modularity, and the effective utilization of technologies to meet the unique demands of educational scheduling. The design phase lays the foundation for a system that not only optimizes timetable creation through advanced algorithms but also provides an intuitive and responsive user experience. Through thoughtful architectural decisions, the Timetable Generation system aims to deliver a reliable solution that adapts to the evolving needs of educational institutions.

6.1. System Architecture:

The system architecture for the Timetable Generation encompasses a multi-tiered structure, integrating front-end, back-end, and database components to ensure a comprehensive and efficient solution. The front-end, developed using technologies like Flutter for cross-platform compatibility, provides a user-friendly interface for administrators, teachers, and students. The back-end, powered by a robust server infrastructure, hosts the scheduling algorithms, WebSocket for real-time updates, and integrates with a database, allowing seamless data retrieval and storage through a chosen Database Management System (DBMS) such as MySQL. This scalable and modular architecture is designed to handle the complexities of educational scheduling, ensuring optimal performance and adaptability to future enhancements. Below is a high-level system architecture for a Timetable Generator based on the specified objectives:

- **User Interface (UI) Layer**
- **Application Logic Layer**
- **Data Management Layer**
- **Security Layer:**
- **Cross-Cutting Concerns Layer**

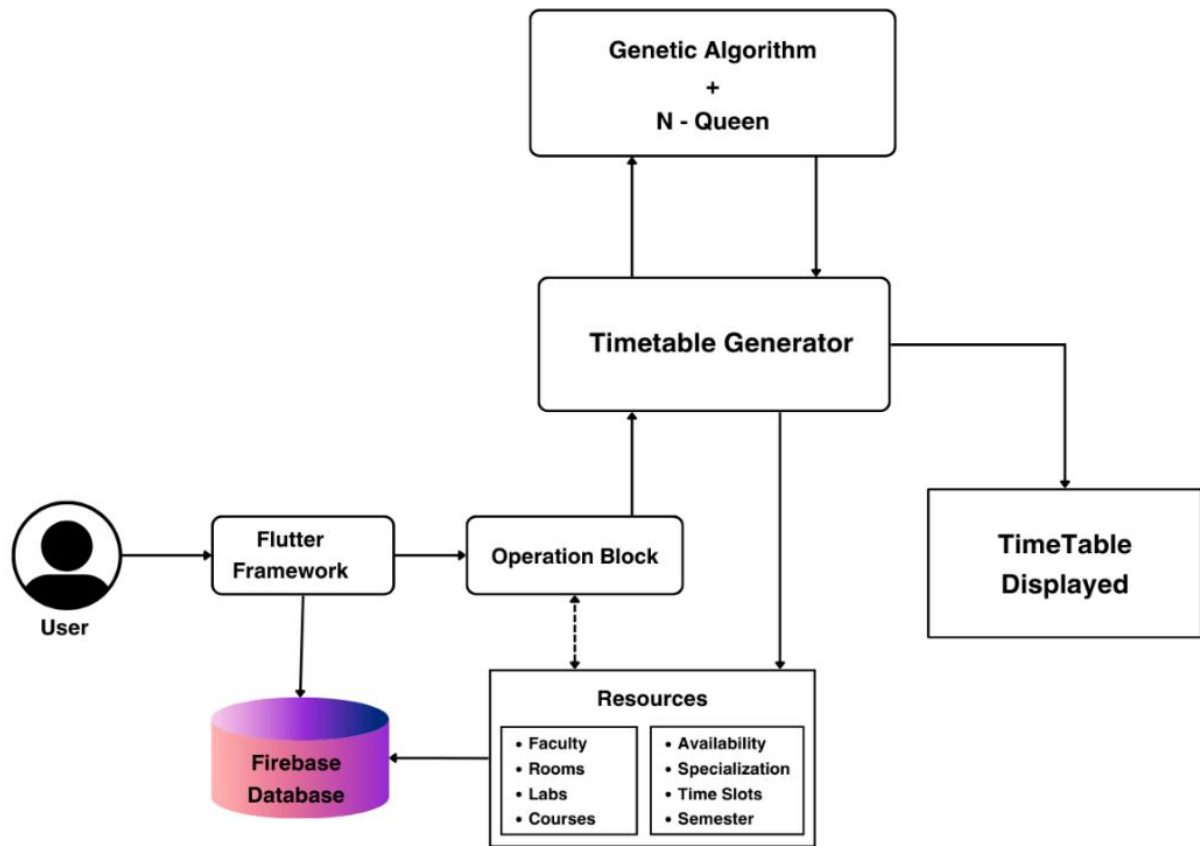


Fig. Proposed System Architecture

6.2. Database Design:

Designing the database for a Timetable Generator involves defining the structure of the database, specifying the tables, relationships, and constraints necessary to store and manage the data efficiently. Below is a detailed database design for the Timetable Generator based on the specified objectives:

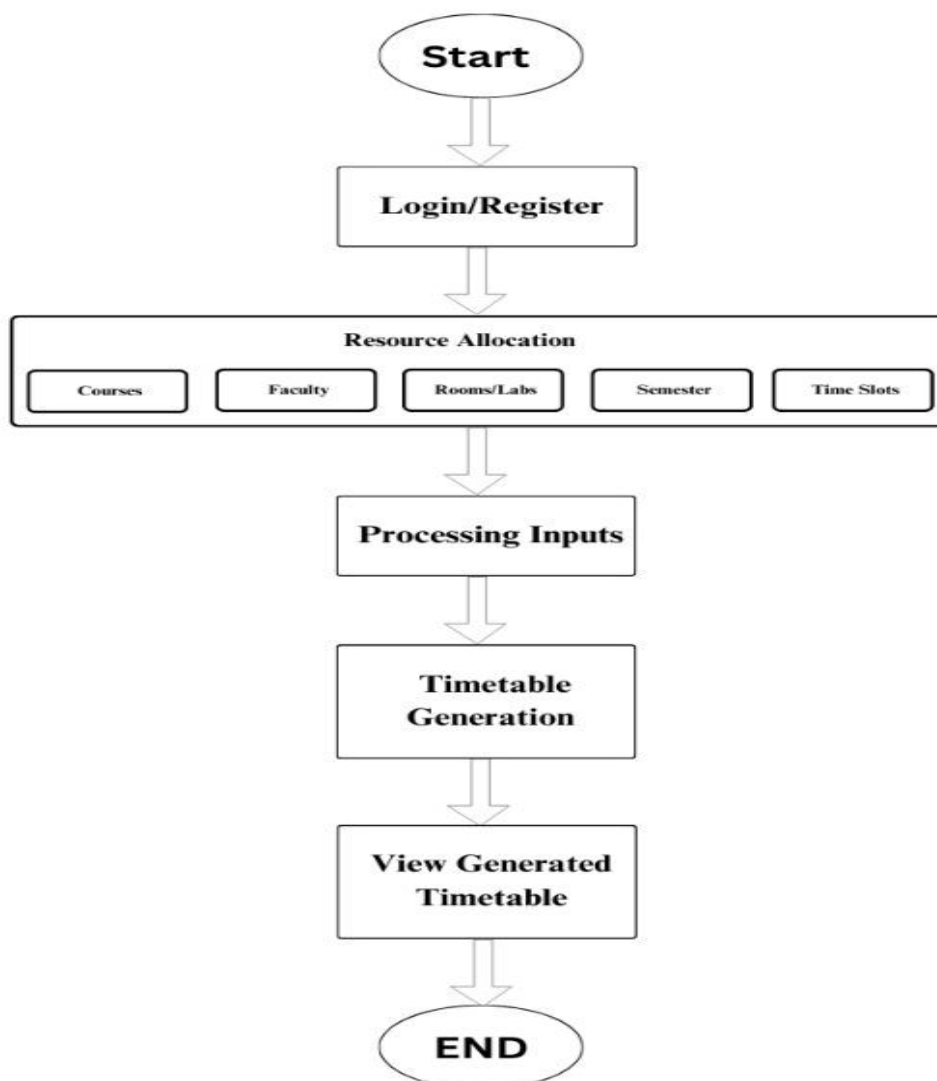
6.3. Advantages of Proposed System:

1. In contrast to traditional manual timetabling methods, the system provides enhanced flexibility.
2. It requires only minimal processing or computing power.

3. Significantly reducing the time required to generate error-free timetables.
4. Facilitating easy data entry and revision through an intuitive interface.
5. Timetables generated typically fall within the range of 60% to 80% optimal solutions.
6. Virtually eliminating the need for paperwork.

6.4. System Design:

System Flowchart:



Flowchart of System

Use case Diagram:

A use-case diagram, in its fundamental display, depicts the interaction between a user and a system while delineating the specifications of a specific usage instance. It visually portrays the various user categories within a system and the varied methods by which they interact with it. Usually, this diagram is utilized in conjunction with textual usage instances and is frequently complemented by other types of diagrams. In this instance, personnel enter semester and subject particulars into the system, which subsequently processes this data to produce the timetable.

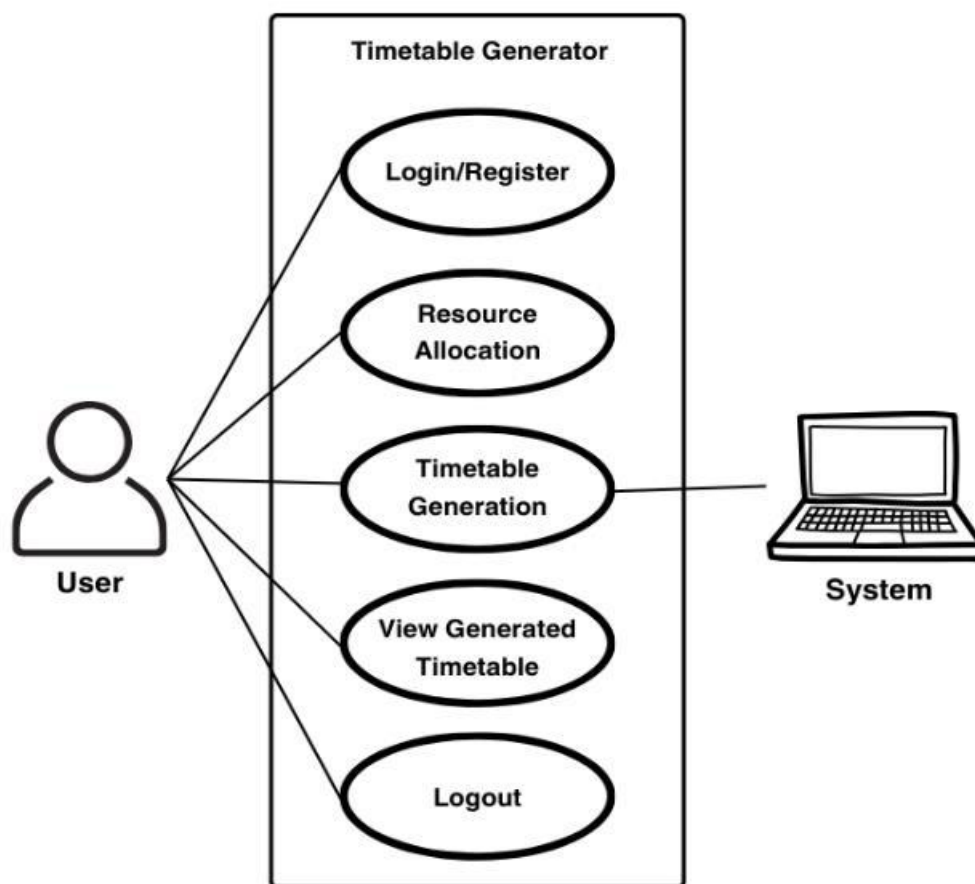


Fig. Use Case Diagram

7. Testing

Introduction to Testing

Testing is an important phase in the development of the “**Time Table Generator**” system, ensuring that all modules work correctly, efficiently, and without errors. The purpose of testing is to identify bugs, validate system performance, and confirm that the application satisfies the user requirements.

Since this project includes multiple functionalities such as **creating a timetable, displaying schedules, searching by day, updating periods, and exiting the program**, testing plays a major role in maintaining the smooth working of the system.

The system was tested under different conditions to ensure **accuracy, reliability, usability, and proper execution of all operations**.

Types of Testing Performed

Functional Testing

Functional testing was performed to verify that each feature of the system works according to the specified requirements.

All modules such as:

- **Create Time Table**
- **Display Time Table**
- **Detailed Time Table**
- **Search by Day**
- **Update Period**
- **Exit Program**

were tested carefully.

Special attention was given to checking whether the data entered by the user is correctly stored in the **2D arrays** and displayed in the required timetable format.

The update functionality was also tested to ensure that the modified subject, teacher name, and room number are correctly replaced.

Integration Testing

Integration testing was carried out to ensure that different parts of the system interact properly with each other.

The communication between:

- **input module**
- **data storage**
- **display module**
- **search module**
- **update module**

was tested thoroughly.

For example, after creating the timetable, the display and search functions were tested to verify that they correctly retrieve the entered data.

Similarly, after updating a period, the display function was tested again to ensure the updated values are shown properly.

User Interface Testing

User interface testing focused on checking the **console-based interaction** of the application.

The menu-driven interface was tested for:

- clarity
- readability
- easy navigation
- correct prompts

The output format of the timetable was checked to ensure it appears in a neat table structure.

Proper messages like:

- **Time Table Created Successfully**
- **Period Updated Successfully**
- **Invalid Choice**

were verified for better user understanding.

Performance Testing

Performance testing was performed to evaluate how efficiently the program behaves with repeated inputs and larger timetable data.

The system was tested by entering timetable details for all **6 days and 6 periods** multiple times.

The program responded efficiently without delay and displayed results quickly under normal conditions.

Since the project is console-based and uses arrays, the performance remained stable and efficient.

Security Testing

Security testing was conducted to ensure safe and controlled execution of the program.

The program was tested against invalid inputs such as:

- wrong menu choices
- invalid day numbers
- invalid period numbers

The system correctly handled invalid choices by displaying appropriate error messages.

This prevents accidental misuse and improves program stability.

Usability Testing

Usability testing was performed to ensure that the application is easy to use and understandable for students, teachers, and administrators.

The menu options were tested with different user scenarios to confirm that users can easily:

- create schedules
- search specific days
- update periods
- view full timetable

The simple structure of the program makes it easy for beginners to understand and operate.

Test Results and Observations

The testing process showed that the system performs efficiently under normal conditions.

All major functionalities, including:

- timetable creation
- display
- search
- update
- exit

worked as expected.

Minor issues such as formatting alignment and invalid input handling were identified and resolved during testing.

The final system was found to be **accurate, reliable, user-friendly, and suitable for academic project submission.**

8. Coding

```
import java.util.Scanner;

public class TimeTableGeneratorProject {

    static String[] days = {

        "Monday", "Tuesday", "Wednesday",

        "Thursday", "Friday", "Saturday"

    };

    static String[][] subjects = new String[6][6];

    static String[][] teachers = new String[6][6];

    static String[][] rooms = new String[6][6];

    static Scanner sc = new Scanner(System.in);

    // Function to create timetable

    public static void createTimeTable() {

        System.out.println("\n===== ENTER TIME TABLE DETAILS =====");

        for (int i = 0; i < days.length; i++) {

            System.out.println("\nEnter details for " + days[i]);
```

```

        for (int j = 0; j < 6; j++) {

            System.out.println("Period " + (j + 1));

            System.out.print("Enter Subject Name: ");

            subjects[i][j] = sc.nextLine();

            System.out.print("Enter Teacher Name: ");

            teachers[i][j] = sc.nextLine();

            System.out.print("Enter Room Number: ");

            rooms[i][j] = sc.nextLine();

        }

    }

    System.out.println("\nTime Table Created Successfully!");

}

// Function to display timetable

public static void displayTimeTable() {

    System.out.println("\n===== WEEKLY TIME TABLE
=====");

    System.out.printf("%-12s %-15s %-15s %-15s %-15s %-15s %-15s\n",

        "Day", "P1", "P2", "P3", "P4", "P5", "P6");

    for (int i = 0; i < days.length; i++) {

```



```

        System.out.printf("%-12s", days[i]);

        for (int j = 0; j < 6; j++) {

            System.out.printf("%-15s", subjects[i][j]);

        }

        System.out.println();

    }

}

// Detailed display with teacher and room

public static void displayDetailedTimeTable() {

    System.out.println("\n===== DETAILED TIME TABLE =====");

    for (int i = 0; i < days.length; i++) {

        System.out.println("\n" + days[i]);

        for (int j = 0; j < 6; j++) {

            System.out.println("Period " + (j + 1));

            System.out.println("Subject : " + subjects[i][j]);

            System.out.println("Teacher : " + teachers[i][j]);

            System.out.println("Room   : " + rooms[i][j]);

            System.out.println("-----");

        }

    }

}

```

```

    }

    // Search by day

    public static void searchByDay() {

        System.out.print("\nEnter day to search: ");

        String searchDay = sc.nextLine();

        for (int i = 0; i < days.length; i++) {

            if (days[i].equalsIgnoreCase(searchDay)) {

                System.out.println("\nTime Table for " + days[i]);

                for (int j = 0; j < 6; j++) {

                    System.out.println("Period " + (j + 1) + " : " +

                        subjects[i][j] + " | " +

                        teachers[i][j] + " | Room " +

                        rooms[i][j]);

                }

                return;

            }

        }

        System.out.println("Day not found!");

    }

```

```

// Update timetable

public static void updatePeriod() {

    System.out.print("\nEnter day number (1-6): ");

    int day = sc.nextInt();


    System.out.print("Enter period number (1-6): ");

    int period = sc.nextInt();

    sc.nextLine();


    System.out.print("Enter new subject: ");

    subjects[day - 1][period - 1] = sc.nextLine();


    System.out.print("Enter new teacher: ");

    teachers[day - 1][period - 1] = sc.nextLine();


    System.out.print("Enter new room: ");

    rooms[day - 1][period - 1] = sc.nextLine();


    System.out.println("Period Updated Successfully!");

}


// Show menu

public static void menu() {

    System.out.println("\n===== TIME TABLE GENERATOR =====");

```

```
System.out.println("1. Create Time Table");  
  
System.out.println("2. Display Time Table");  
  
System.out.println("3. Detailed Time Table");  
  
System.out.println("4. Search by Day");  
  
System.out.println("5. Update Period");  
  
System.out.println("6. Exit");  
  
System.out.print("Enter your choice: ");  
  
}
```

```
public static void main(String[] args) {
```

```
    int choice;
```

```
    do {
```

```
        menu();
```

```
        choice = sc.nextInt();
```

```
        sc.nextLine();
```

```
        switch (choice) {
```

```
            case 1:
```

```
                createTimeTable();
```

```
                break;
```

```
            case 2:
```

```
                displayTimeTable();
```

```
                break;
```

```
        case 3:

            displayDetailedTimeTable();

            break;

        case 4:

            searchByDay();

            break;

        case 5:

            updatePeriod();

            break;

        case 6:

            System.out.println("Exiting Program...");

            break;

        default:

            System.out.println("Invalid Choice!");

    }

} while (choice != 6);

sc.close();

}

}
```

Output

1.Create Time Table Output

```
1. Create Time Table
2. Display Time Table
3. Detailed Time Table
4. Search by Day
5. Update Period
6. Exit
Enter your choice: 1

Enter details for Monday
Period 1
Enter Subject Name: Java
Enter Teacher Name: Sharma Sir
Enter Room Number: 101

Period 2
Enter Subject Name: DBMS
Enter Teacher Name: Verma Sir
Enter Room Number: 102

Period 3
Enter Subject Name: OS
Enter Teacher Name: Gupta Sir
Enter Room Number: 103

Period 4
Enter Subject Name: Maths
Enter Teacher Name: Khan Sir
Enter Room Number: 104

Period 5
Enter Subject Name: EVS
Enter Teacher Name: Singh Sir
Enter Room Number: 105

Period 6
Enter Subject Name: Python
Enter Teacher Name: Mishra Sir
Enter Room Number: 106

Time Table Created Successfully!
```

2.Display time table output

Enter your choice: 2

Day	P1	P2	P3	P4	P5	P6
Monday	Java	DBMS	OS	Maths	EVS	Python
Tuesday	C++	Java	DBMS	CN	Maths	Lab
Wednesday	Python	OS	Java	EVS	DBMS	Lab
Thursday	Maths	Java	C++	Python	CN	OS
Friday	DBMS	EVS	Java	Lab	Maths	Python
Saturday	Seminar	Project	Viva	Java	Lab	Free

3.Detailed time table output

Enter your choice: 3

Monday

Period 1

Subject : Java

Teacher : Sharma Sir

Room : 101

Period 2

Subject : DBMS

Teacher : Verma Sir

Room : 102

Period 3

Subject : OS

Teacher : Gupta Sir

Room : 103

4.Search by day output

```
Enter your choice: 4
Enter day to search: Monday

Time Table for Monday
Period 1 : Java | Sharma Sir | Room 101
Period 2 : DBMS | Verma Sir | Room 102
Period 3 : OS | Gupta Sir | Room 103
Period 4 : Maths | Khan Sir | Room 104
Period 5 : EVS | Singh Sir | Room 105
Period 6 : Python | Mishra Sir | Room 106
```

5.Update period output

```
Enter your choice: 5

Enter day number (1-6): 1
Enter period number (1-6): 5

Enter new subject: Data Science
Enter new teacher: Chamoli Sir
Enter new room: 201

Period Updated Successfully!
```

After updating,if you display again

Monday	Java	DBMS	OS	Maths	Data Science	Python
--------	------	------	----	-------	--------------	--------

6.Exit output

```
Enter your choice: 6
Exiting Program...
```


9. Reports

- The Time Table Generator system provides reports for daily and weekly schedules.
- It displays subjects, teachers, room numbers, and period timings in a structured format.
- The report helps students and teachers easily view their class schedules.
- It also generates updated reports whenever any period is modified.
- The system ensures accurate timetable management and quick access to information.
- Summary reports include total working days, total periods, and assigned subjects.
- These reports improve schedule organization and reduce manual effort.
- Overall, the reporting module makes timetable management simple and efficient.

10. Future Scope

The **Time Table Generator** project has significant scope for future enhancement and development. At present, the system is a **console-based Java application** designed to create, display, search, and update weekly schedules.

In the future, this project can be extended into a more advanced and user-friendly system with additional features.

One major enhancement can be the development of a **Graphical User Interface (GUI)** using **Java Swing or JavaFX**, which will make the application more interactive and easier to use.

The project can also be upgraded into a **database-based system** by integrating **MySQL / Oracle Database**, where timetable records can be permanently stored and retrieved whenever required.

Another future improvement is the implementation of an **automatic timetable generation algorithm**, where subjects, teachers, rooms, and periods are assigned automatically without conflicts. This would reduce manual work and save time.

The system can further be expanded into a **web-based or mobile application** for schools, colleges, and universities so that students and teachers can access the timetable anytime and anywhere.

11. Bibliography

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- 4.**Head First Java** – Kathy Sierra & Bert Bates ([Reddit](#))
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Websites Referred

- 1.**Oracle Java Documentation** – Official Java reference and tutorials ([Oracle](#))
- 2.**GeeksforGeeks** – Java programs and concepts
- 3.**W3Schools Java Tutorial** – basic syntax and examples
- 4.**TutorialsPoint Java** – theory and coding examples

Software Tools Used

- 1.**Visual Studio Code** - Used for coding and development.
- 2.**Google chrome** - Used for testing, debugging and inspecting UI elements.
- 3.**GitHub** - Used for version control and project management(if applicable).